Momentum in two dimensions

antocciphysics | TEACHER KEY | CPA

Learning target: Conserve momentum separately along perpendicular axes.

Suggested first pass: 1, 2, 3, 7. Add independent, mixed and stretch work from the bank as time and student evidence permit. Estimated times are planning estimates, not completion requirements.

01. Bring back an earlier idea (retrieve) *Review: Unit 3, Day 7; 3 course days later.*
East positive: 5-3=2 kg·m/s east. Opposite directions require opposite signs.

02. Explain a worked example (worked)
Right triangle with legs 3 and 4; resultant 5 northeast.

03. Look and explain (model)
Result points northeast; ordinary lengths are not diagonal magnitude.

04. Build the solution (practice)
3 m/s east and 4 m/s north.

05. Use the idea (reason)
Otherwise the two equations may describe different interactions or omit parts of the system; conservation comparison becomes inconsistent.

06. Choose a method (mixed) *Review: Unit 2, Day 14; 12 course days later.*
Fastest at 4 AU. Sweeping equal areas in equal times requires greater speed near the Sun.

07. Exit check (exit)
2 m/s west and 1 m/s south.

08. STRETCH: Apply it another way (stretch)
Components 3 m/s east and 4 m/s north; supplied 3-4-5 relationship gives speed 5 m/s.

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01. Bring back an earlier idea (retrieve) *Review: Unit 3, Day 7; 3 course days later.*
Deviation≈3.333%. It compares values relative to the initial value; uncertainty requires a measurement/error analysis.

02. Explain a worked example (worked)
Right triangle with legs 3 and 4; resultant 5 northeast.

03. Represent and interpret (model)
Result points northeast; ordinary lengths are not diagonal magnitude.

04. Solve and justify (practice)
Components (2, 4/3) m/s; speed 2.40 m/s; angle 33.7°.

05. Analyze the reasoning (reason)
Otherwise the two equations may describe different interactions or omit parts of the system; conservation comparison becomes inconsistent.

06. Choose a method (mixed) *Review: Unit 2, Day 14; 12 course days later.*
$T_B/T_A=4^{(3/2)}\approx8$. The normalization depends on the central mass and chosen units.

07. Exit check (exit)
(6,-3) kg·m/s;√45=6.71 kg·m/s.

08. STRETCH: Apply it another way (stretch)
v=(2, 4/3) m/s; speed≈2.40 m/s; 33.7° north of east.